

(h) Estimation

Students should be taught to:

estimate population means from samples;

estimate of population proportions from samples with application in opinion polls and elsewhere;

understand the effect of sample size on estimates and the variability of estimates.

3 Reasoning, interpreting and discussing results

Content

Students should be taught in the context of real data to:

apply statistical reasoning, explain and justify inferences, deductions, arguments and solutions;

explore connections, look for and examine relationships between variables;

consider the limitations of any assumptions;

relate summarised data to any initial questions or observations;

interpret all forms of statistical tables, diagrams and graphs;

compare distributions of data and make comparisons using measures of central tendency, measures of dispersion and percentiles;

Notes

Cases clearly restricted to the content of the specification at the appropriate level.

Eg height and weight, age and depreciation of a car, GNP and mortality in infants.

Simple cases only, eg honest replies to questionnaires, equally likely outcomes in probabilities, representativeness of sample of population, reliability of secondary data.

The relevance of measures of central tendency.

To include real published tables and graphs.

The shapes of distributions and graphs may be used. Formula for variance and standard deviation to be given.

Content

check results for reasonableness and modify their approaches if necessary;

interpret correlation as a measure of the strength of the association between two variables.

4 Probability

Content

Students should be taught to:

understand the meaning of the words event and outcome;

understand words such as impossible, certain, highly likely, likely, unlikely, possible, evens and present these on a likelihood scale;

put outcomes in order in terms of probability;

put probabilities in order on a probability scale;

understand the terms 'random' and 'equally likely';

understand and use measures of probability from a theoretical perspective and from a limiting frequency or experimental approach;

understand that in some cases the measure of probability based on limiting frequency is the only viable measure;

compare expected frequencies and actual frequencies;

use probability to assess risk;

produce, understand and use a sample space;

understand the terms mutually exclusive and exhaustive and to understand the addition law $P(A \text{ or } B) = P(A) + P(B)$;

Notes

Eg the mean must lie between the maximum and minimum, 'the average bicycle speed was 130 km per hour' is not reasonable.

The use of words such as weak or strong will be expected.

Notes

Tossing a coin is an event with outcomes landing Heads or Tails.

Interpretation of real-life situations will be expected, eg 'the probability that the horse will win the next race is 0.3'; 'the probability that I will get a grade C or better in my Statistics is $\frac{3}{4}$.'

Use of $\leq$ will be expected.

Labelling of the scale will be expected.

Formal definition and notation of a limit will not be required whilst terminology such as 'as the number of trials increases' will be required.

The probability of a sports team winning can only be measured from a limiting frequency perspective. For example, medical statistics for assessment of health risks.

Examples may be taken from insurance.

Listing all outcomes of single events and two successive events, in a systematic way.

$P(A \text{ or } B) = P(A) + P(B)$;

'Mutually exclusive' means that the occurrence of one outcome prevents another, $\sum p = 1$ when summed over all mutually exclusive outcomes.

Content

know, for mutually exclusive outcomes, that the sum of the probabilities is one and in particular the probability of something not happening is one minus the probability of it happening;

form and use tree diagrams and probability tree diagrams for independent events;

understand, use and apply the addition law for mutually exclusive events and the multiplication law for independent events.

Notes

If $P(A) = p$ then $P(\text{not } A) = 1 - p$.

Listing all possible joint or compound outcomes.

To correctly apply

$$P(A \text{ and } B) = P(A) \times P(B),$$

$$P(A \text{ or } B) = P(A) + P(B).$$